

Apache Kafka Training — 32 Hours

Reorganized Course Outline (Grouped by Theme)

This outline preserves all original topics, modules, hours, and exercises from the source curriculum, but regroups them into clearer thematic tracks so related concepts flow together.

Track 1: Kafka Foundations

Understanding messaging systems and Kafka's core architecture.

1.1 Messaging Systems & Introduction to Kafka

- Introduction to Messaging Systems and their use cases; problems and limitations (P2P, PubSub)
- Publish-subscribe messaging and its role in the big data ecosystem — JMS, RabbitMQ, ActiveMQ
- Kafka for Big Data & Data Ingestion; role in ETL
- Why Kafka is needed; components of Kafka
- Kafka features and use cases
- Installing Kafka - Single node
- **Exercise:** *Single Node Single Broker Configuration*

1.2 Kafka Architecture

- Kafka architecture overview
- Components of a cluster and design
- Role of Zookeeper
- Core concepts: Broker, Topics, Partitions, Producer, Consumer, Leader, Follower
- Replication in Kafka, Offsets, ISR (In-Sync Replicas)
- Consumer Groups
- **Exercise:** *Exploring CLI Commands*

Track 2: Cluster Setup & Management

Deploying and managing Kafka clusters across environments.

2.1 Managing Clusters

Zookeeper

- Zookeeper Cluster concepts
- **Exercise:** *Exploring Zookeeper*

Multi Node kafka Cluster

- Kafka Cluster
- Multiple nodes – multiple broker clusters (EC2 Instances)
- **Exercise:** *Multiple Brokers Configuration*

Track 3: Producer & Consumer Development

Building Java applications that read from and write to Kafka.

3.1 Producer and Consumer API

- Writing Producers (sync and async)
- Writing Consumers
- **Exercise:** *Text Producer and Consumer*
- Custom Partitioner and Custom Serializer
- Single Consumer vs. Multithreaded Consumer (scaling consumers)
- Automatic partition reassignment and consumer-group rebalancing
- **Exercise:** *Partitioner, Serializer*

Track 4: Topic, Partition & Broker Configuration

Tuning and operating topics, partitions, and brokers at a deeper level.

4.1 Topic and Partition Management — 4 Hours

- Compacted Topics and Compression concepts
- Broker, Producer, and Consumer configurations:
 - acks, buffer.memory, compression.type, retries, batch.size, linger.ms
 - client.id, max.in.flight.requests.per.connection, timeout.ms, metadata.fetch.timeout.ms
- Partition management & configuration:
 - num.partitions, log.retention.ms, log.retention.bytes, log.segment.bytes, log.segment.ms, message.max.bytes
- Deleting specific records from a topic
- Deleting all records from a topic
- Deleting a topic; Disaster Recovery (DR) and Mirror Maker
- Message retention and backpressure/consumer lag
- **Exercise:** *Compacted Topics*
- Kafka Log Directories
- Kafka Reassign Partitions
- **Exercise:** *Kafka More CLI Concepts.*

Track 5: Security & Operations

Securing Kafka and operating it reliably in production.

5.1 Security & Administration

- Kafka administration tools
- Frequent problems and solutions
- Tuning brokers, OS, producers, consumers; cross-data-center communication; best practices
- Kafka Manager usage

- Kafka configuration files — details and troubleshooting
- Kafka Security overview
- Understanding Authorization and Access Control Lists (ACL)
- **Exercise: ACL**
- Wire Encryption using SSL
- **Exercise: SSL**

(Avro Producer/Consumer is covered once, in depth, under Track 8 alongside Schema Registry.)

Track 6: Data Integration with Kafka Connect

Building pipelines to move data in and out of Kafka.

6.1 Data Pipelines with Kafka Connect

- Kafka Connect overview and architecture
- Implementing a pipeline
- Common use cases
- File Connectors, JDBC Connectors, Cassandra Connectors
- **Exercise: JDBC Connect**
- **Exercise: Elastic Search Connectors**

Track 7: Stream Processing

Processing data in motion using Kafka Streams and ksqlDB.

7.1 Kafka Stream Processing and KSQL Server

- Stream processing concepts
- Kafka Streams architecture
- Choosing a stream processing framework
- Using KStream and KTable
- KSQLDB Servers
- Creating Stream, Table, and Query using KSQL server and Control Center; REST API
- Terminating and monitoring Stream, Table, and joining KStream with KTable
- **Exercise: KSQL DB**

Track 8: Schema Management & Governance

Managing data contracts and monitoring the Kafka ecosystem.

8.1 Schema Registry and Confluent Control Center

- Schema Registry overview; schema as a first-class citizen
- Sample: Avro schema
- Drawbacks; other formats
- Confluent Schema Registry as a safeguard

- Operations and terminology; on Windows
- Validate operations
- Add/Delete Schema; Subjects; List all Subjects
- Avro Producer and Consumer
- Compatibilities
- UI
- ***Exercise: AVRO Producer and Consumer***

Summary of Hours by Track

Track / Topic Area	Hours	
Track 1 — Kafka Foundations	6	
Track 2 — Cluster Setup & Management	2	
Track 3 — Producer & Consumer Development	4	
Track 4 — Topic, Partition & Broker Configuration	4	
Track 5 — Security & Operations	4	
Track 6 — Data Integration with Kafka Connect	4	
Track 7 — Stream Processing	4	
Track 8 — Schema Management & Governance	4	
Total	32	

Note: The original "Day" structure (Day 1–4) has been replaced with theme-based tracks. If a day-by-day delivery schedule is still needed, Tracks 1–2 map to Day 1, Tracks 3–4 to Day 2, Tracks 5–6 to Day 3, and Tracks 7–8 to Day 4 — matching the original hour allocations exactly.